

DBMS LAB ASSIGNMENT - 1

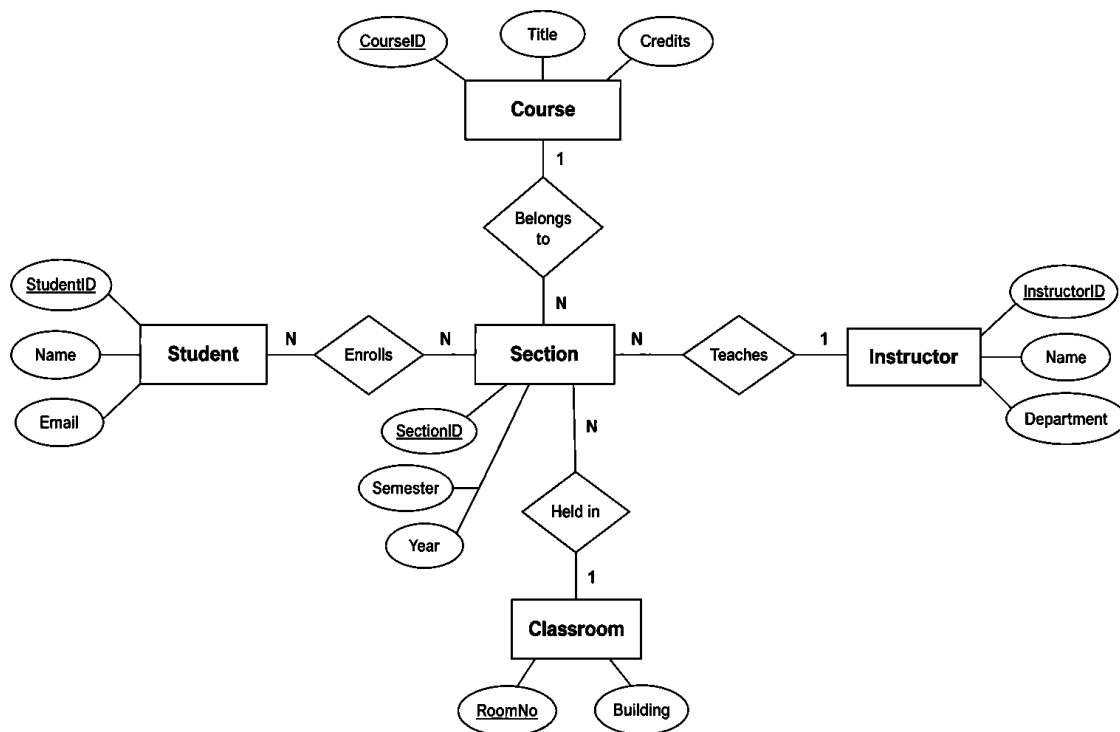
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ER Diagram – 1

Flaws:

According to the given rules, a course can have different sections and each section can have a separate instructor. Therefore, a separate entity named “Section” is required. Students should enroll in a section, and each course can have multiple sections.

1. Semester and Year should be attributes of Section because the same course can be offered in different sections, semesters, and years. If these are stored only with Course, information about different offerings may be lost.
2. A Student should enroll in a Section rather than directly in a Course because the student registers for a particular section.
3. Since different sections of the same Course may be taught by different instructors, the Instructor should teach a Section rather than the Course.
4. Classroom allocation can vary between sections, so a particular Section should be held in a Classroom rather than the Course itself.

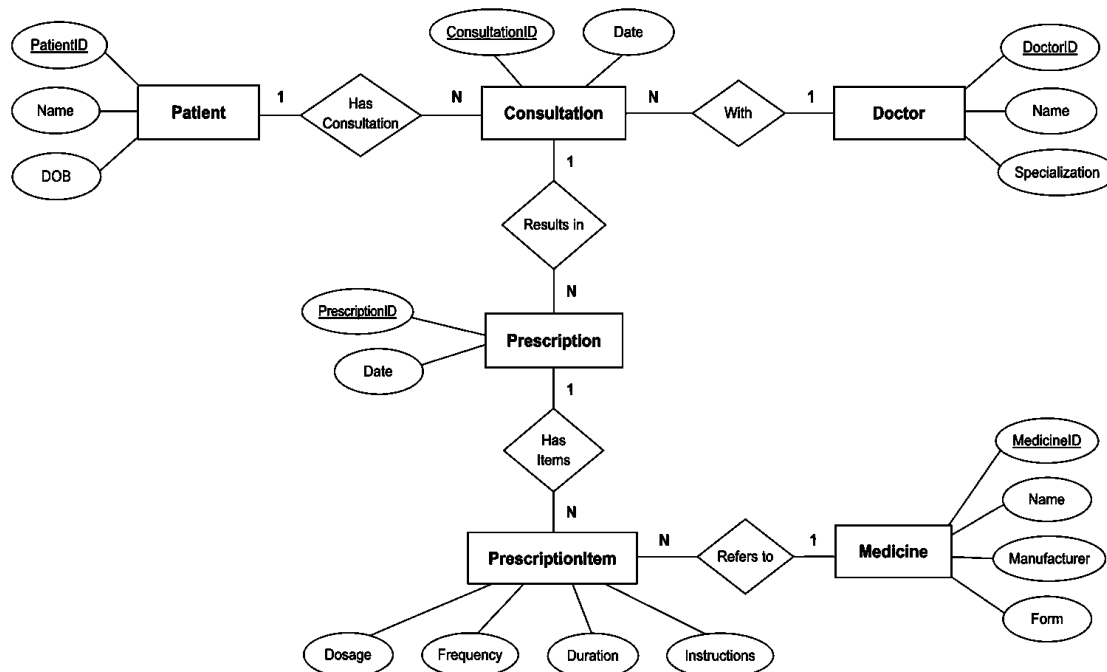


ER Diagram – 2

Flaws:

To preserve consultation history, a separate “Consultation” entity is needed because a patient may visit the same doctor many times. A consultation can produce a “Prescription”, and each prescription can contain “PrescriptionItem” records because dosage belongs to the prescription item, not to the Medicine.

1. Consultation should be an entity because the same patient may consult the same doctor multiple times. This allows the consultation history to be maintained.
2. Prescription should also be an entity so that the prescription history for each consultation can be stored.
3. PrescriptionItem should be an entity because the same medicine may be given to different patients with different dosages.
4. For time-dependent instructions, Frequency and Duration can be stored as attributes of PrescriptionItem.

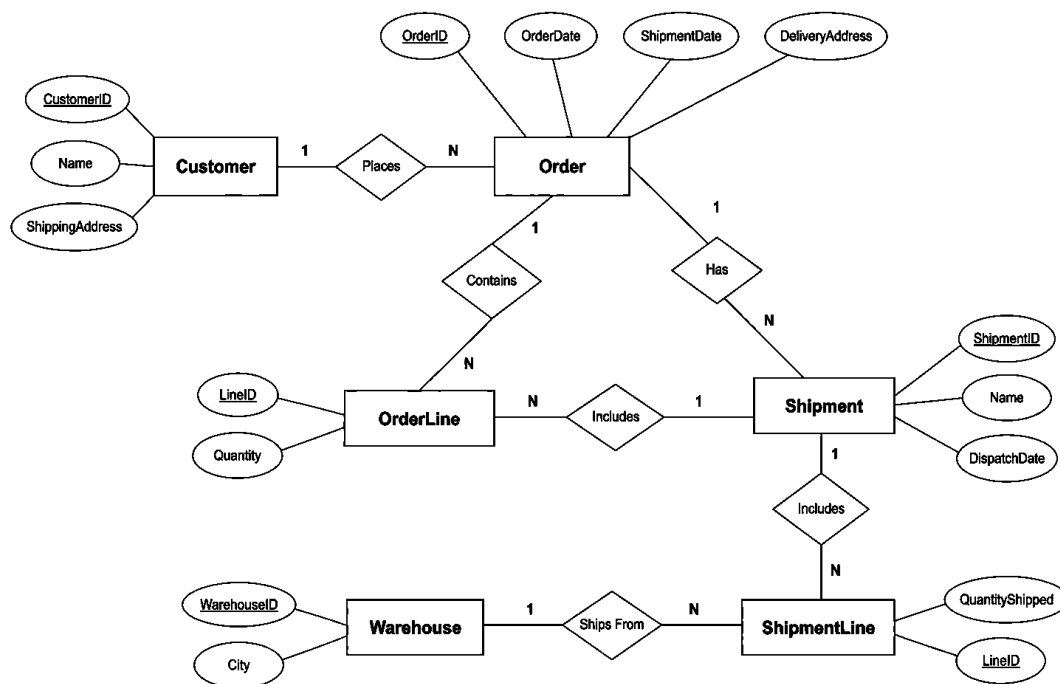


ER Diagram – 3

Flaws:

The “OrderLine” entity is missing even though quantity is specific to an order line rather than the Product. A “ShipmentLine” entity is also required to represent partial and split shipments correctly.

1. Address should be an attribute of Order rather than Customer because a customer may use different delivery addresses for different orders.
2. An OrderLine entity is required because quantity belongs to the OrderLine and not to the Product.
3. A ShipmentLine entity is required to keep track of partially shipped OrderLines.
4. Shipment should have its own Dispatch Date so that an order can be fulfilled through multiple shipments.

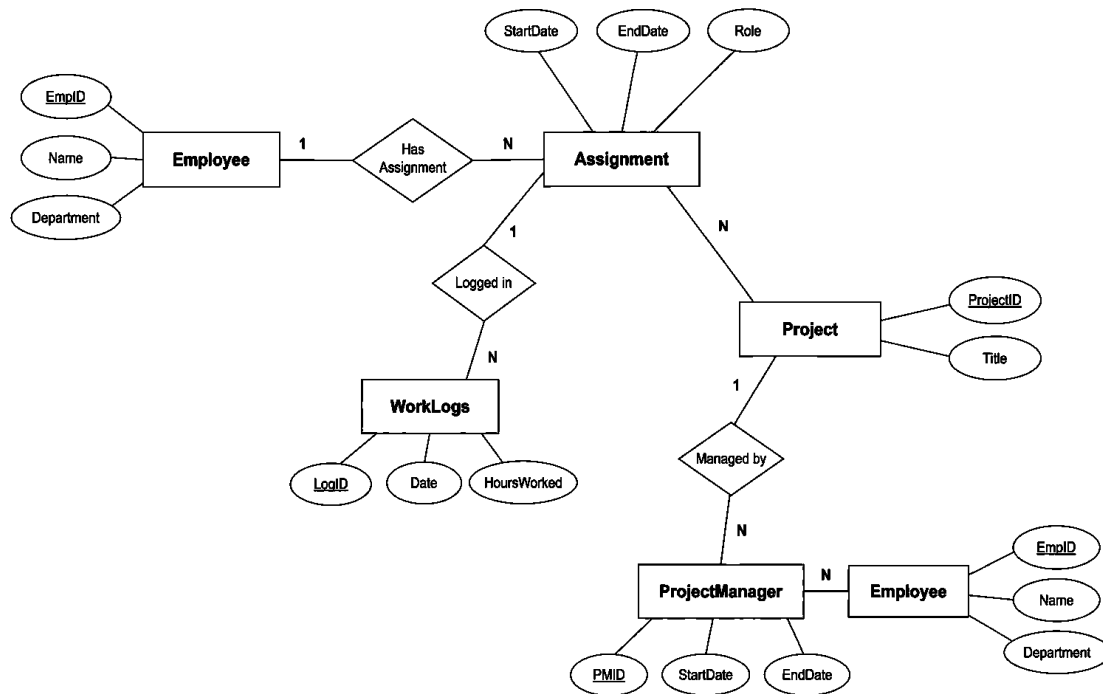


ER Diagram – 4

Flaws:

The relationship between Employee and Project should be represented through an entity so that the Employee–Project history can be maintained properly.

1. There should be an Assignment entity to store the role of a particular employee in a project.
2. The Works_On relationship cannot represent an employee joining, leaving, and later rejoining a project. Therefore, Assignment should have Start Date and End Date.
3. Hours are recorded for an employee-project assignment over a time period, so they should not be stored directly in Project. A WorkLogs entity can store the hours worked for each time period and assignment.
4. Start Date and End Date should be associated with the MANAGED_BY relationship because project managers may change over time and the management history needs to be retained.



ER Diagram – 5

Flaws:

The current ER model does not properly represent bookings, flight occurrences, aircraft changes, and seat assignments.

1. A PNR entity is required because one PNR can contain multiple passengers and multiple flight segments.
2. A FlightOccurrence entity is required because the same flight number can operate on different dates.
3. Aircraft should be connected to FlightOccurrence because the aircraft assigned to a flight may change for different occurrences.
4. A SeatAssignment entity is required to store different seats for the same passenger on different flight segments.

